DGMS 2019 GEOLOGY • QUESTION 40

Crystallography: Standard Notation of Miller Indices for Crystal Faces

Topic: Geometric Crystallography & Lattice Planes | Primary Reference: Dexter Perkins (Mineralogy, 3rd ed., Ch. 10 & 11)

Question Statement

Q.40. Which one of the following expressions is correct as per notations of Miller indices of a crystal face?

(a) (1 1 1) [or (1 1 0)]
(b) (½ ½ 1)
(c) (1a 1b 1c)
(d) 1 1 1

a (1 1 1) (Correct Answer)	b (½ ½ 1)
c (1a 1b 1c)	d 1 1 1

✓ Correct Answer: (a) (1 1 1) [Miller indices of a crystal face are integers cleared of common fractions, written without axis letters, and strictly enclosed in round parentheses: (h k l)].

Textbook Evidence: Dexter Perkins (Chapter 11)

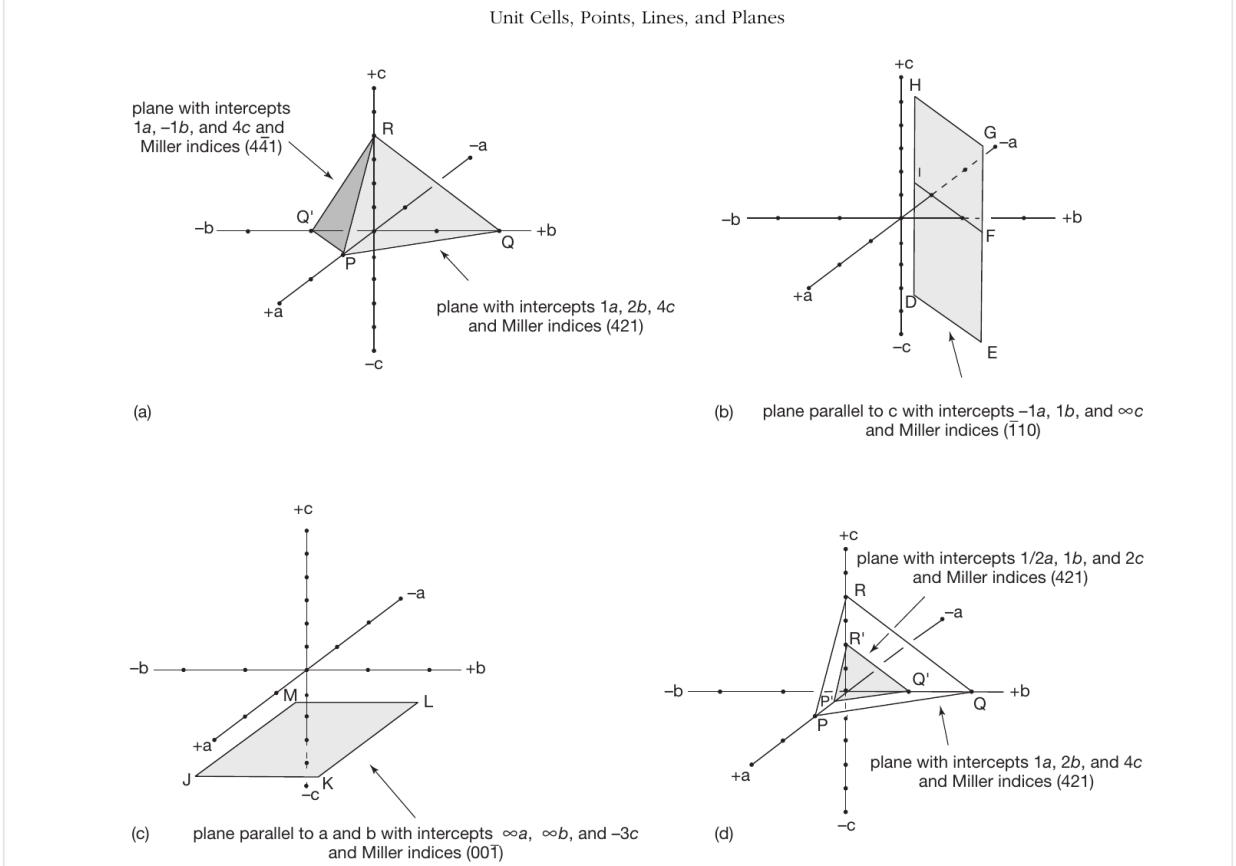


FIGURE 10 Planes intersecting three, two, or one axis: (a) planes intersecting all three axes; (b) a plane parallel to the *c* axis; (c) a plane parallel to two axes; (d) two parallel planes intersecting all three axes.

make generalizations. The index (*hk*0) describes the family of faces with their third index equal to zero. A

The relationship between planes and directions in a crystal depends on the crystal system. Except in

Figure 11.10: Geometry of Miller Indices for Crystal Faces (Dexter Perkins, *Mineralogy*, 3rd ed., p. 320). Derivation of Miller indices from axial intercepts: (a) A plane intersecting crystallographic axes at intercepts $1a, 2b, 4c$. Taking reciprocals ($1/1, 1/2, 1/4$) and clearing fractions by multiplying by 4 yields the whole-number Miller index **(421)**. (b) A plane parallel to the c-axis with intercepts $-1a, 1b, \infty c$ yields the index **(1$\bar{1}$0)**. All crystal faces are designated by whole-number integers enclosed in parentheses without commas or axis letters.

> *"To determine the Miller index of a crystal face:*
> *1. Determine the intercepts of the face on the crystallographic axes $a, b,$ and c in terms of unit dimensions.*
> *2. Invert the intercepts (take their reciprocals). A face parallel to an axis intersects it at infinity (∞), yielding a reciprocal of 0.*
> *3. Clear fractions to reduce the reciprocals to the smallest whole integers h, k, l.*
> *4. Enclose the numbers in parentheses:* **$(h k l)$**. *Negative indices are denoted by a bar over the number (e.g., $\bar{1}$).*"
>
> **— Dexter Perkins, *Mineralogy*, Chapter 11 (Unit Cells, Points, Lines, and Planes), pp. 319–322**

Comprehensive Option Analysis

(a) (1 1 1) **[CORRECT]**

Crystallographic Rules of Miller Indices: By international crystallographic convention (established by William Hallowes Miller in 1839):

- A **crystal face** is denoted strictly by enclosing the whole integers $(h k l)$ within **round parentheses**, with no commas and no crystallographic axis letters.
- For example, the octahedron face in the isometric system has intercepts $1a, 1b, 1c \rightarrow$ reciprocals $1, 1, 1 \rightarrow$ written as **(1 1 1)** or **(111)**.
- The expression strictly follows every single rule of Miller index notation.

(b) (½ ½ 1) **[INCORRECT]**

Miller indices **must be whole integers**. Fractional notations like $(1/2, 1/2, 1)$ represent fractional intercepts or raw reciprocals before clearing fractions. By rule, fractions must be cleared by multiplying by the least common denominator (which would convert $(1/2, 1/2, 1)$ into the valid Miller index $(1 1 2)$).

(c) (1a 1b 1c) **[INCORRECT]**

Expressions incorporating the crystallographic axis labels (such as $1a : 1b : 1c$) are **Weiss parameters (axial parameters)**, not Miller indices. Miller indices are dimensionless integers (h, k, l) and never contain the letters $a, b,$ or c.

(d) 1 1 1 (without parentheses) **[INCORRECT]**

In crystallography, the type of brackets used defines the physical entity being described. Bare numbers without parentheses fail standard crystallographic syntax and do not specify a crystal face.

Standard Crystallographic Bracket Notation Guide

Bracket Type	Example	Geometrical Entity Represented	Physical Significance
Parentheses: $(h k l)$	**(1 1 1) or (1 0 0)**	**Individual Crystal Face or Lattice Plane**	A single planar surface of a crystal or specific family of parallel atomic lattice planes.
Curly Braces: $\{h k l\}$	$\{1 0 0\}$ or $\{1 1 1\}$	**Crystal Form**	A complete group of symmetry-equivalent crystal faces (e.g., $\{100\}$ represents all 6 cube faces).
Square Brackets: $[u v w]$	$[0 0 1]$ or $[1 1 0]$	**Zone Axis** / **Direction Vector**	A linear vector or zone axis pointing along a lattice line (e.g., $[001]$ points along the c-axis).
Angle Brackets: $\langle u v w \rangle$	$\langle 1 0 0 \rangle$	**Family of Symmetry-Equivalent Directions**	All crystallographically equivalent direction vectors in the lattice.

Key Takeaway & Exam Strategy

Universal Crystallographic Bracket Rules:

- **Face:** $(h k l)$ $\rightarrow$ Parentheses, whole numbers only, no commas, no axis letters.
- **Form:** $\{h k l\}$ $\rightarrow$ Braces.
- **Zone / Direction:** $[u v w]$ $\rightarrow$ Square brackets.
- **Weiss Parameters:** Intercepts with axis letters (e.g., $1a : \infty b : 2c$), which are converted to Miller indices by taking reciprocals and clearing fractions.